

Econometrics I

Lecture 12a: Maximum Likelihood

Chris Conlon

NYU Stern

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Consider a linear regression with $\varepsilon_i \mid X_i \sim \mathcal{N}(0, \sigma^2)$

$$Y_i = X_i \beta + \varepsilon_i$$

We've discussed the **least squares estimator**:

$$\hat{\beta}_{ols} = \arg \min_{\beta} \sum_{i=1}^N (Y_i - X_i \beta)^2$$

$$\hat{\beta}_{ols} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Review: What is a Likelihood?

Suppose we write down the joint distribution of our data (y_i, x_i) for $i = 1, \dots, n$.

$$\mathbb{P}(y_1, \dots, y_n, x_1, \dots, x_n \mid \theta)$$

If (y_i, x_i) are I.I.D then we can write this as:

$$\mathbb{P}(y_1, \dots, y_n, x_1, \dots, x_n \mid \theta) = \prod_{i=1}^N \mathbb{P}(y_i, x_i \mid \theta) \propto \prod_{i=1}^N \mathbb{P}(y_i \mid x_i, \theta) = \mathbb{L}(\mathbf{y} \mid \mathbf{x}, \theta)$$

(the $\propto$ step requires that the distribution of x_i does not depend on θ). We call this $\mathbb{L}(\mathbf{y} \mid \mathbf{x}, \theta)$ the **likelihood** of the observed data.

MLE: Example

If we know the distribution of ε_i we can construct a **maximum likelihood estimator**

$$(\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2) = \arg \max_{\beta, \sigma^2} \mathbb{L}(\beta, \sigma^2)$$

Where

$$\begin{aligned}\mathbb{L}(\beta, \sigma^2) &= \prod_{i=1}^N \mathbb{P}(y_i \mid x_i, \beta, \sigma^2) \\ &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2\sigma^2} (Y_i - X_i\beta)^2 \right] \\ \ell(\beta, \sigma^2) &= -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (Y_i - X_i\beta)^2\end{aligned}$$

Take the FOC's of: $\ell(\beta, \sigma^2) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (Y_i - X_i\beta)^2$:

$$\frac{\partial \ell(\beta, \sigma^2)}{\partial \beta} = \frac{1}{\sigma^2} \sum_{i=1}^N X_i' (Y_i - X_i\beta) = 0 \rightarrow \hat{\beta}_{MLE} = \hat{\beta}_{OLS}$$

$$\frac{\partial \ell(\beta, \sigma^2)}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^N (Y_i - X_i\beta)^2 = 0$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{N} \sum_{i=1}^N (Y_i - X_i\hat{\beta}_{MLE})^2$$

Note: the unbiased estimator uses $\frac{1}{N-K}$ (K parameters, including the intercept).

MLE: General Case

1. Start with the **joint density of the data** $Z_1, \dots, Z_N$ with density $f_Z(z, \theta)$
2. Construct the likelihood function of the sample $z_1, \dots, z_n$

$$\mathbb{L}(\mathbf{z} \mid \theta) = \prod_{i=1}^N f_Z(z_i, \theta)$$

3. Construct the **log likelihood** (this has the same arg max)

$$\ell(\mathbf{z} \mid \theta) = \sum_{i=1}^N \ln f_Z(z_i, \theta)$$

4. Take the FOC's to find $\hat{\theta}_{MLE}$ that solves

$$\frac{\partial \ell(\theta)}{\partial \theta} = 0 \text{ or } \nabla_{\theta} \ell(\theta) = 0 \text{ or } \sum_{i=1}^N \frac{\partial \ln f_Z(z_i, \theta)}{\partial \theta} = 0.$$

Basic Setup: we know the form of $F(z \mid \theta)$ but not θ_0 . We know $\theta_0 \in \Theta \subset \mathbb{R}^K$.

- ▶ Begin with a sample of z_i from $i = 1, \dots, N$ which are I.I.D. with CDF $F(z|\theta_0)$.
- ▶ The MLE chooses

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \ell(\theta) = \arg \max_{\theta} \sum_{i=1}^N \ln f_Z(z_i, \theta)$$

- ▶ In practice the computer typically **minimizes negative log-likelihood** so that $\hat{\theta}_{MLE} = \arg \min_{\theta} -\ell(\theta)$.

1. Consistency. When is it true that, for every $\varepsilon > 0$,

$$\lim_{N \rightarrow \infty} \mathbb{P} \left(\left\| \hat{\theta}_{MLE} - \theta_0 \right\| > \varepsilon \right) = 0$$

2. Asymptotic Normality. What else do we need to show?

$$\sqrt{N} \left(\hat{\theta}_{MLE} - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, -\mathbb{E} \left[\frac{\partial^2 \log f(Z_i, \theta_0)}{\partial \theta \partial \theta'} \right]^{-1} \right)$$

3. Optimization. How do we obtain $\hat{\theta}_{MLE}$ anyway?

- ▶ $Z_i \sim \mathcal{N}(\theta_0, 1)$ and $\Theta = (-\infty, \infty)$. In this case:

$$\ell(\theta) = -\frac{N}{2} \ln(2\pi) - \sum_{i=1}^N (z_i - \theta)^2 / 2$$

- ▶ MLE is $\hat{\theta}_{MLE} = \bar{z}$ which is consistent for $\theta_0 = \mathbb{E}[Z_i]$
- ▶ Asymptotic distribution is $\sqrt{N}(\bar{z} - \theta_0) \sim \mathcal{N}(0, 1)$.
- ▶ Calculating mean is easy!

- ▶ $Z_i = (Y_i, X_i)$ where X_i has finite mean and variance (but arbitrary distribution)
- ▶ $(Y_i|X_i = x) \sim \mathcal{N}(x'\beta_0, \sigma_0^2)$

$$\hat{\beta}_{MLE} = (X'X)^{-1}X'Y$$
$$\hat{\sigma}_{MLE}^2 = \frac{1}{N} \sum (y_i - x_i'\hat{\beta}_{MLE})^2$$

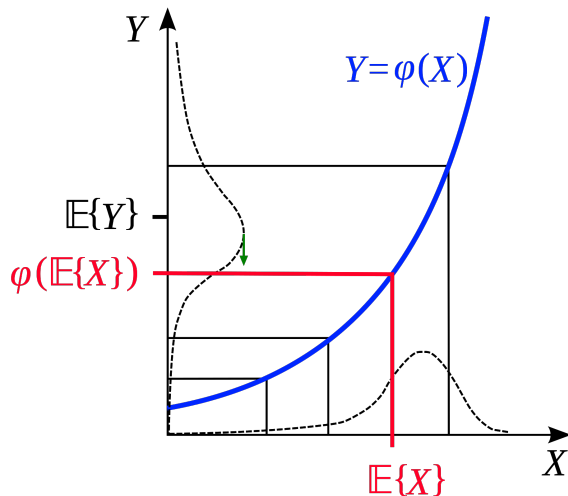
- ▶ We already have shown consistency and AN for linear regression with normally distributed errors...

- ▶ $Z_i = (Y_i, X_i)$ where X_i has finite mean and variance (but arbitrary distribution)
- ▶ $\mathbb{P}(Y_i = 1|X_i = x) = \frac{e^{x'\theta_0}}{1+e^{x'\theta_0}}$
- ▶ Solution is the **logit** model.
- ▶ No simple MLE solution, establishing properties is not obvious...

Jensen's Inequality

Let $\phi(z)$ be a convex function.

Then $\mathbb{E}[\phi(Z)] \geq \phi(\mathbb{E}[Z])$, with strict inequality if ϕ is strictly convex and Z is non-degenerate.



Define Y as the ratio of the density at θ to the density at the true value θ_0 both evaluated at Z

$$Y = \frac{f_Z(Z; \theta)}{f_Z(Z; \theta_0)}$$

- ▶ Let $g(a) = -\ln(a)$ so that $g'(a) = \frac{-1}{a}$ and $g''(a) = \frac{1}{a^2}$.
- ▶ Then by **Jensen's Inequality** $\mathbb{E}[-\ln Y] \geq -\ln \mathbb{E}[Y]$.
- ▶ This gives us

$$\mathbb{E}_z \left[-\ln \left(\frac{f_Z(Z; \theta)}{f_Z(Z; \theta_0)} \right) \right] \geq -\ln \left(\mathbb{E}_z \left[\frac{f_Z(Z; \theta)}{f_Z(Z; \theta_0)} \right] \right)$$

- ▶ The RHS is

$$\mathbb{E}_z \left[\frac{f_Z(Z; \theta)}{f_Z(Z; \theta_0)} \right] = \int \frac{f_Z(z; \theta)}{f_Z(z; \theta_0)} \cdot f_Z(z; \theta_0) dz = \int f_Z(z; \theta) dz = 1$$

Because $\log(1) = 0$ this implies:

$$\mathbb{E}_z \left[-\ln \left(\frac{f_Z(Z; \theta)}{f_Z(Z; \theta_0)} \right) \right] \geq 0$$

Therefore

$$-\mathbb{E} [\ln f_Z(Z; \theta)] + \mathbb{E} [\ln f_Z(Z; \theta_0)] \geq 0$$

$$\mathbb{E} [\ln f_Z(Z; \theta_0)] \geq \mathbb{E} [\ln f_Z(Z; \theta)]$$

- ▶ We maximize the expected value of the log likelihood at the true value of θ !
- ▶ Helpful to work with $\mathbb{E}[\log f(z; \theta)]$ sometimes.

We can relate the **Fisher Information** to the Hessian of the log-likelihood

$$\mathcal{I}(\theta_0) = -\mathbb{E} \left[\frac{\partial^2 \ln f}{\partial \theta \partial \theta'} (z; \theta_0) \right] = \mathbb{E} \left[\frac{\partial \ln f}{\partial \theta} (z; \theta_0) \times \frac{\partial \ln f}{\partial \theta} (z; \theta_0)' \right]$$

- ▶ This is sometimes known as the **outer product of scores**.
- ▶ This matrix is **positive semi-definite** (positive definite under identification)
- ▶ Recall that $\mathbb{E} \left[\frac{\partial \ln f}{\partial \theta} (z; \theta_0) \right] = 0$ at the true θ_0 (and the sample mean score is 0 at $\hat{\theta}_{MLE}$)

$$1 = \int_z f_Z(z; \theta) dz \Rightarrow 0 = \frac{\partial}{\partial \theta} \int_z f_Z(z; \theta) dz$$

With some regularity conditions

$$0 = \int_z \frac{\partial f_Z}{\partial \theta}(z; \theta) dz = \underbrace{\int_z \frac{\partial \ln f_Z}{\partial \theta}(z; \theta) \cdot f_Z(z; \theta) dz}_{\mathbb{E}_\theta \left[\frac{\partial \ln f_Z}{\partial \theta}(Z; \theta) \right]}$$

- At $\theta = \theta_0$: the score has mean zero, which is the population version of the FOC.
- Can get information identity with another set of derivatives.

The Cramér–Rao Bound

We can relate the **Fisher Information** to the Hessian of the log-likelihood

$$\mathcal{J}(\theta) = -\mathbb{E} \left[\frac{\partial^2 \ln f}{\partial \theta \partial \theta'} (Z|\theta) \right]$$

It turns out this provides a bound on the variance of any **unbiased** estimator from N observations:

$$\text{Var}(\hat{\theta}) \geq [N \mathcal{J}(\theta_0)]^{-1}$$

Since $\sqrt{N}(\hat{\theta}_{MLE} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{J}(\theta_0)^{-1})$, the MLE attains the bound asymptotically: it is **asymptotically efficient** (if the model is correctly specified).

Tradeoffs

- ▶ How does this compare to GM Theorem?
- ▶ If MLE is the most efficient estimator, why ever use something else?

We are going to practice writing down the:

- ▶ Likelihood $L(\theta) = \mathbb{P}(z_1, \dots, z_n; \theta) = \prod_{i=1}^N f(z_i; \theta)$.
- ▶ log likelihood $\ell(\theta) = \sum_{i=1}^N \ln f(z_i; \theta) = \sum_{i=1}^N \ell_i(z_i; \theta)$.
- ▶ Scores $\mathcal{S}_i(z_i, \theta) = \frac{\partial \ln f(z_i; \theta)}{\partial \theta} = \frac{\partial \ell_i(z_i; \theta)}{\partial \theta}$.
- ▶ Hessian contribution $\mathcal{H}_i(z_i, \theta) = \frac{\partial^2 \ell_i(z_i; \theta)}{\partial \theta \partial \theta'}$.
- ▶ Information Matrix $\mathcal{I}(\theta) = \mathbb{E}_z[-\mathcal{H}_i(z_i, \theta)] = \mathbb{E}_z[\mathcal{S}_i(z_i, \theta) \mathcal{S}_i(z_i, \theta)^T]$
- ▶ Variance of any unbiased estimator $\geq [N \mathcal{I}(\theta)]^{-1}$ (Cramér–Rao Lower Bound).

- ▶ Suppose we have data that are exponentially distributed $Y_i \sim \text{Exp}(\lambda)$. The goal is to estimate the parameter λ via MLE and $f(y_i|\lambda) = \lambda e^{-\lambda y_i}$ so that $\mathbb{E}[y_i] = \frac{1}{\lambda}$.
- ▶ Sometimes we want to parameterize the rate $\lambda_i = x_i' \beta$ with covariates (this requires $x_i' \beta > 0$; $\lambda_i = \exp(x_i' \beta)$ is the usual choice).
- ▶ Example: Time until next customer arrives varies with time of day, day of week, etc. Time until default might depend on credit score, debt-to-income, market conditions, etc.

Exponential Regression Example (Careful: two parameterizations here!)

$$\text{For } y_i > 0, \quad f_{y|x}(y_i | x_i, \beta) = (x_i \beta) e^{-(x_i \beta) y_i} \quad \text{so that } \mathbb{E}[y_i] = \frac{1}{x_i \beta}$$

With log likelihood

$$\ell(\beta) = \sum_{i=1}^N \ln f_{y|x}(y_i | x_i, \beta) = \sum_{i=1}^N \log(x'_i \beta) - (x'_i \beta) y_i$$

And Score, Hessian, and Information Matrix:

$$\begin{aligned} \frac{\partial \ell_i}{\partial \beta} &= \mathcal{S}_i(y_i, x_i, \beta) = x'_i \left(\frac{1}{x_i \beta} - y_i \right) \\ \frac{\partial^2 \ell_i}{\partial \beta \partial \beta'} &= \mathcal{H}_i(y_i, x_i, \beta) = -\frac{x'_i x_i}{(x'_i \beta)^2} \\ \mathcal{J}(\beta_0) &= \sum_i \frac{x'_i x_i}{(x_i \beta)^2} \end{aligned}$$

Computing Maximum Likelihood Estimators

Newton's Method for Root Finding

Consider the Taylor series for $f(x)$ approximated around x_0 :

$$f(x) \approx f(x_0) + f'(x_0) \cdot (x - x_0) + \frac{1}{2}f''(x_0) \cdot (x - x_0)^2 + O((x - x_0)^3)$$

Suppose we wanted to find a **root** of the equation where $f(x^*) = 0$ and solve for x :

$$\begin{aligned} 0 &= f(x_0) + f'(x_0) \cdot (x - x_0) \\ x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \end{aligned}$$

This gives us an **iterative** scheme to find x^* :

1. Start with some x_0 . Calculate $f(x_k), f'(x_k)$
2. Update using $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$
3. Stop when $|x_{k+1} - x_k| < \epsilon_{tol}$.

Newton-Raphson for Optimization

We can re-write **optimization** as **root finding**:

- ▶ We want to know $\hat{\theta} = \arg \max_{\theta} \ell(\theta)$.
- ▶ Construct the FOCs $\frac{\partial \ell}{\partial \theta} = 0$ and find the zeros.
- ▶ How? using Newton's method! Set $f(\theta) = \frac{\partial \ell}{\partial \theta}$

$$\theta_{k+1} = \theta_k - \left[\frac{\partial^2 \ell}{\partial \theta^2}(\theta_k) \right]^{-1} \cdot \frac{\partial \ell}{\partial \theta}(\theta_k)$$

The SOC (for a maximum) is that $\frac{\partial^2 \ell}{\partial \theta^2} < 0$. Ideally at all θ_k .

This is all for a **single variable** but the **multivariate** version is basically the same.

Start with the objective $Q(\theta) = -\ell(\theta)$:

- ▶ Approximate $Q(\theta)$ around some initial guess $\theta^{(0)}$ with a quadratic function
- ▶ Minimize the quadratic function (because that is easy) call that $\theta^{(1)}$
- ▶ Update the approximation and repeat.

$$\theta_{k+1} = \theta_k - \left[\frac{\partial^2 Q}{\partial \theta \partial \theta'}(\theta_k) \right]^{-1} \frac{\partial Q}{\partial \theta}(\theta_k)$$

- ▶ The equivalent SOC is that the Hessian Matrix of Q is **positive definite** (ideally at all θ).
- ▶ In that case Q is **strictly convex**, so any stationary point is its **unique global minimum** (the maximum of ℓ) and is easy to find.

We can generalize to Quasi-Newton methods:

$$\theta_{k+1} = \theta_k - \lambda_k \underbrace{\left[\frac{\partial^2 Q}{\partial \theta \partial \theta'} \right]^{-1}}_{A_k} \frac{\partial Q}{\partial \theta}(\theta_k)$$

Two Choices:

- ▶ Step length λ_k
- ▶ Step direction $d_k = A_k \frac{\partial Q}{\partial \theta}(\theta_k)$
- ▶ Often rescale the direction to be unit length $\frac{d_k}{\|d_k\|}$.
- ▶ If we use A_k as the true Hessian and $\lambda_k = 1$ this is a **full Newton step**.

Newton's Method: Alternatives

Choices for A_k

- ▶ $A_k = I_k$ (Identity) is known as **gradient descent** or **steepest descent**
- ▶ BHHH. Specific to MLE. Exploits the **Fisher Information**.

$$A_k = \left[\sum_{i=1}^N \frac{\partial \ln f}{\partial \theta} (z_i, \theta_k) \frac{\partial \ln f}{\partial \theta'} (z_i, \theta_k) \right]^{-1}, \quad \text{because}$$

$$\frac{1}{N} \sum_{i=1}^N \frac{\partial \ln f}{\partial \theta} \frac{\partial \ln f}{\partial \theta'} \xrightarrow{p} \mathbb{E} \left[\frac{\partial \ln f}{\partial \theta} (Z, \theta_0) \frac{\partial \ln f}{\partial \theta'} (Z, \theta_0) \right] = -\mathbb{E} \left[\frac{\partial^2 \ln f}{\partial \theta \partial \theta'} (Z, \theta_0) \right]$$

- ▶ Quasi-Newton alternatives **BFGS** (the usual default), **SR1**, and **DFP** start from an initial estimate of the Hessian matrix and then cheaply update A_k .
- ▶ Usually computing (and inverting) the Hessian is the costly step.
- ▶ Non invertible Hessians are bad news.

Thanks!
